

Precautionary Saving in Europe: Feedback Implementation

Liquidity, Term Structure, Risk Premia, and a Forward-Looking Lens

Project documentation — feedback extension

June 29, 2026

Abstract

This addendum implements supervisor feedback on the savings analysis. Its organising idea is a single methodological correction: stop treating *cash and deposits* as the only precautionary buffer, and instead rank household assets on a *liquidity / maturity ladder*. On that ladder, the post-2022 move *out of* instant-access cash — which the project’s “follow-the-money” extension had read as evidence of *yield-chasing rather than precaution* — looks very different: the money did not leave the liquid buffer, it moved up the risk/return scale while staying sellable within days. We then add the four further pieces the feedback asked for: the money supply divided by term ($M1/M2/M3$); observable risk premia set against geopolitical tension; a forward-looking expectations composite; and an energy-price liquidity-adequacy analysis; debt securities are split by term for a finer ladder, and a United States comparison (Fed Z.1) sets the euro-area picture in relief. All series are free (Eurostat, the ECB Data Portal, FRED, the Caldara–Iacoviello GPR index, and the US Financial Accounts); the scripts reuse the main project’s helpers. Headline result: the broad sellable-fast share of household saving flows *rose* from 54% to 73% after 2022, and about half of household financial wealth is sellable within days — so the precaution-versus-yield dichotomy is false. Households’ forward-looking income expectations predict the saving rate far better than the geopolitical index does, and the buffer is concentrated up the income distribution, leaving the most energy-exposed households uncushioned. Figures come from a clean live run on June 29, 2026; they are revised by the providers and should be regenerated before citation.

Contents

1	The question, revisited	2
2	Data and method	2
3	The liquidity ladder	3
4	Money supply, divided by term	5
5	Risk premia and geopolitical tension	6
6	A forward-looking lens	7
7	Energy prices and the adequacy of cash	8
8	United States comparison	9

9 What this says about the question	10
10 Limitations	10

1 The question, revisited

Where the main project left off. The project tests whether the euro area’s elevated household saving rate is *substantially precautionary* — a response to uncertainty rather than to income, wealth, or the interest rate alone. The time-series leg was supportive; the cross-country and formal identification legs were inconclusive. The sharpest negative result came from the “follow-the-money” extension: after 2022 the share of saving going into instant-access cash collapsed while money flowed into bonds and term deposits. Read through a binary — *instant-access cash = precautionary, bonds and term deposits = yield-chasing* — this looked like evidence *against* precaution.

The feedback. The supervisors’ objection targets exactly that binary. Many assets can be sold fast, so cash and deposits are not the only precautionary store of value; we should look harder at each asset type and at the money supply, *divided by term*; examine risk premia and whether they rise with geopolitical tension; adopt a more forward-looking approach; and discuss the risk of holding too little cash to absorb rising energy prices — in short, *when* households save, *where*, and how *liquid* those holdings are.

The organising reframe. A bond, a listed share, or a money-market fund unit is sellable in days. Treating it as “locked” understates the precautionary buffer. We therefore replace the binary with a four-rung liquidity / maturity ladder (Table 1); we divide finer where the data allow — debt securities are split *by term*, short-term (F31) into near-money and long-term (F32) into the marketable tier — and recompute the composition of saving on it. The monetary aggregates are the central bank’s own liquidity-by-term tiering of the money-holding sector’s claims, so they are the natural macro counterpart. The remaining sections add the risk-premium, forward-looking, and energy dimensions, and Section 9 states what all of this implies for the precautionary question.

2 Data and method

Every script is standalone, reuses the shared helpers in `_common.py` (lifted from the main project’s modules), and pulls only free, keyless data, writing figures to `figures/` and a report plus CSVs to `data/`. Table 2 lists the additional series. Two helpers are new: a generalised `household_instruments()` that runs the *same* tier classification on flows (`NASA_10_F_TR`) and on stocks (`NASA_10_F_BS`); and `ecb_sdmx()`, a keyless pull from the ECB Data Portal for the monetary aggregates.

A guard worth stating up front: Eurostat does not always publish the F5 sub-instruments (F511/F512/F519/F521/F522). The classifier auto-detects what is present and falls back *conservatively* — an un-splittable equity / funds block is assigned to the illiquid tier T4 — so every “broad-liquid” figure below is a *lower bound*. We never overstate liquidity. In the run reported here the MMF split (F521) was unavailable, so fund shares enter T3 whole.

Table 1: The liquidity / maturity ladder (ESA 2010 instrument codes).

Tier	Instruments	Rationale
T1 — instant ($\approx M1$)	F21 currency, F22 overnight deposits	settle immediately, no price risk
T2 — near-money ($\approx M2/M3$)	F29 other deposits (term & redeemable at notice), F521 MMF shares, F31 short-term debt securities	accessible at short notice / maturity, little duration risk
T3 — marketable	F32 long-term debt securities , F511 listed shares, F522 non-MMF investment-fund shares	sellable within days at market price (duration / price risk)
T4 — illiquid / contractual	F512, F519 unlisted & other equity; F6 insurance, pension & guarantee schemes	locked, contractual, or with no ready secondary market

Table 2: Additional data series (all free / keyless).

Series	Source	Used in
Household financial balance sheet (stocks)	Eurostat NASA_10_F_BS	§3
Household financial transactions (flows)	Eurostat NASA_10_F_TR	§3
Monetary aggregates $M1, M2, M3$	ECB Data Portal (BSI); FRED fall-back	§4
Euro high-yield credit spread (OAS)	FRED BAMLHE00EHYIOAS	§5
Italy & Germany 10y yields	FRED IRLTLT01ITM156N, IRLTLT01DEM156N	§5
Implied equity volatility (VIX)	FRED VIXCLS	§5, §6
Consumer survey (saving, unemployment)	Eurostat EI_BSCO_M	§6
Energy HICP	Eurostat PRC_HICP_MIDX / MANR (NRG)	§7

3 The liquidity ladder

Stocks: what households hold. Re-ranking the household financial balance sheet (Figure 1, Table 3), instant cash (T1) is only about a fifth of financial wealth, but the *broad sellable-fast* share (T1+T2+T3) is around *half*. The remainder (T4 $\approx 47\%$) is dominated by insurance and pension claims and by unlisted business equity — genuinely illiquid. The point the feedback asked for is immediate: the mobilisable buffer is far larger than cash.

Flows: where the marginal saving went. The reframe is starkest in the flows (Figure 2, Table 4). The share of yearly financial saving going into instant cash did collapse, from 66% (2015–19) to 2% (2022+) — the headline of the old binary. But the *broad sellable-fast* share *rose*, from 54% to 73%: the money left cash and went into term deposits, money-market funds, bonds and listed shares, every one of which can be sold within days. On the ladder this is not “leaving the precautionary buffer”; it is holding the buffer in a higher-yielding but still-liquid form once positive rates made cash costly to hold.

This is the single most important result for the precautionary question: it removes the

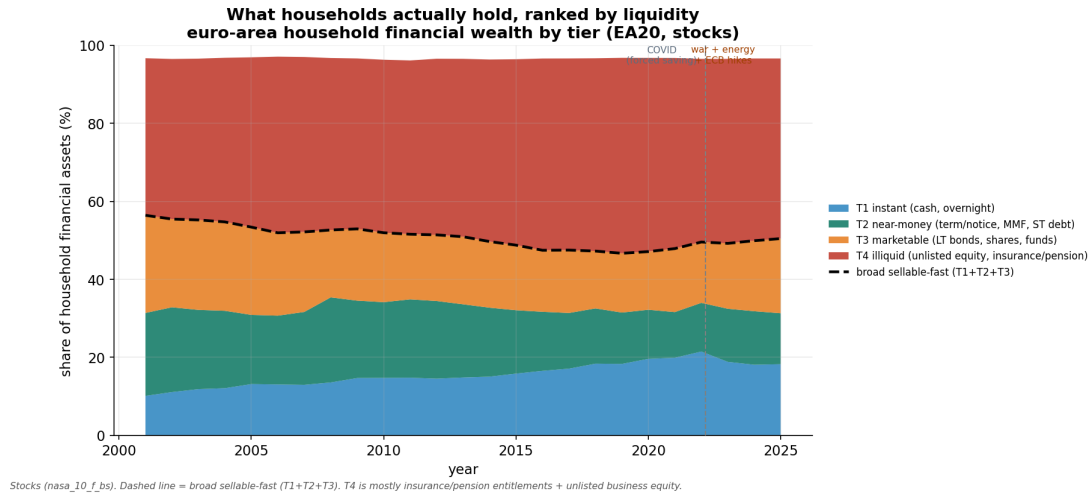


Figure 1: Euro-area household financial wealth by liquidity tier (stocks, NASA_10_F_BS). Only ~ 18% is instant cash (T1); the dashed line — the broad sellable-fast share T1+T2+T3 — sits near 50%.

Table 3: Stock shares of household financial wealth (%), pre- vs post-shock averages.

	2015–19	2022+	change (pp)
T1 instant (cash, overnight)	17.2	19.2	+1.9
T2 near-money (term/notice, MMF)	14.5	13.0	–1.5
T3 marketable (bonds, shares, funds)	15.8	17.6	+1.9
T4 illiquid (unlisted eq., ins./pension)	49.1	46.8	–2.3
<i>narrow cash (T1)</i>	17.2	19.2	+1.9
<i>broad sellable-fast (T1+T2+T3)</i>	47.5	49.8	+2.3

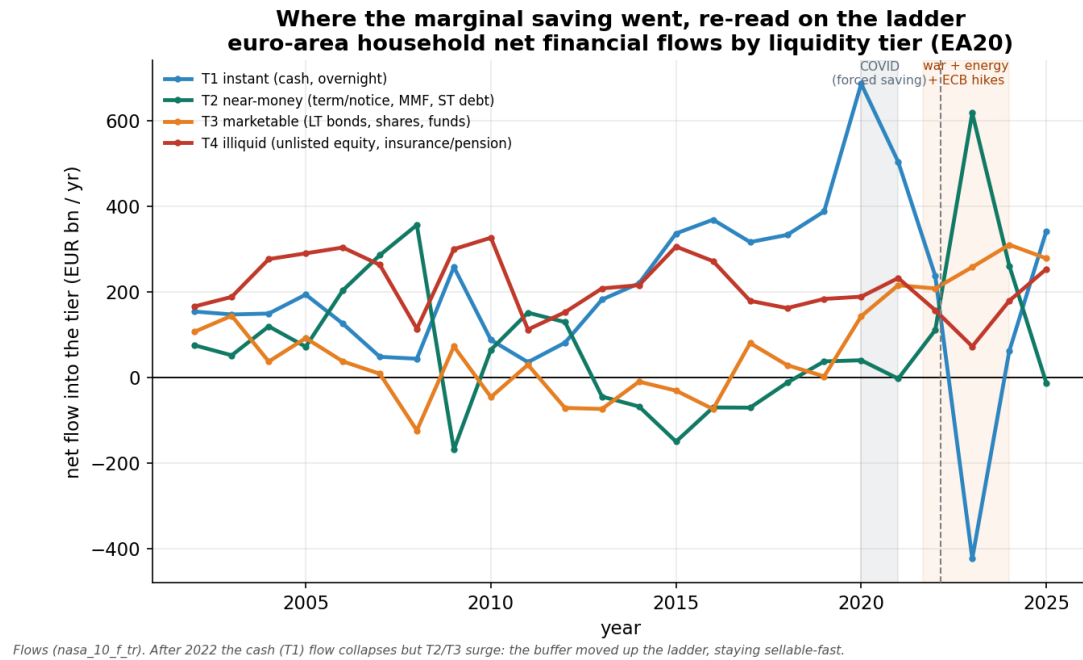


Figure 2: Euro-area household net financial flows by liquidity tier (NASA_10_F_TR). The instant-cash line collapses after 2022 while the marketable tier (T3) surges — a move *within* the liquid part of the ladder.

Table 4: Flow shares of yearly financial saving (%), pre- vs post-shock averages. (Net-flow shares can exceed 100% or go negative; read the levels in Figure 2 alongside.)

	2015–19	2022+	change (pp)
T1 instant (cash, overnight)	66.2	2.3	–63.9
T2 near-money (term/notice, MMF)	–10.7	33.4	+44.2
T3 marketable (bonds, shares, funds)	–1.0	37.4	+38.4
T4 illiquid (unlisted eq., ins./pension)	43.0	20.7	–22.3
<i>narrow cash (T1)</i>	66.2	2.3	–63.9
<i>broad sellable-fast (T1+T2+T3)</i>	54.4	73.0	+18.6

strongest piece of evidence *against* the hypothesis. “Out of cash” is not “out of the buffer.” It does not, by itself, *prove* the motive is precaution — staying liquid is consistent with precaution, not proof of it — but it restores precaution as a live reading after the binary had seemingly ruled it out.

4 Money supply, divided by term

The monetary aggregates tell the same story at the macro level (Figure 3, Table 5). As the ECB raised rates, money migrated *out of* instant overnight balances ($M1$ fell by about EUR 1.2tn between 2022 and 2024) and *into* term and notice deposits ($M2 - M1$ rose from EUR 3.4tn to EUR 4.9tn) and marketable near-money ($M3 - M2$: MMF shares, repo, short debt). Broad money ($M3$) kept growing and co-moves with the household saving rate (annual correlation +0.27). Money itself is tiered by term, and the buffer is not just $M1$ cash.

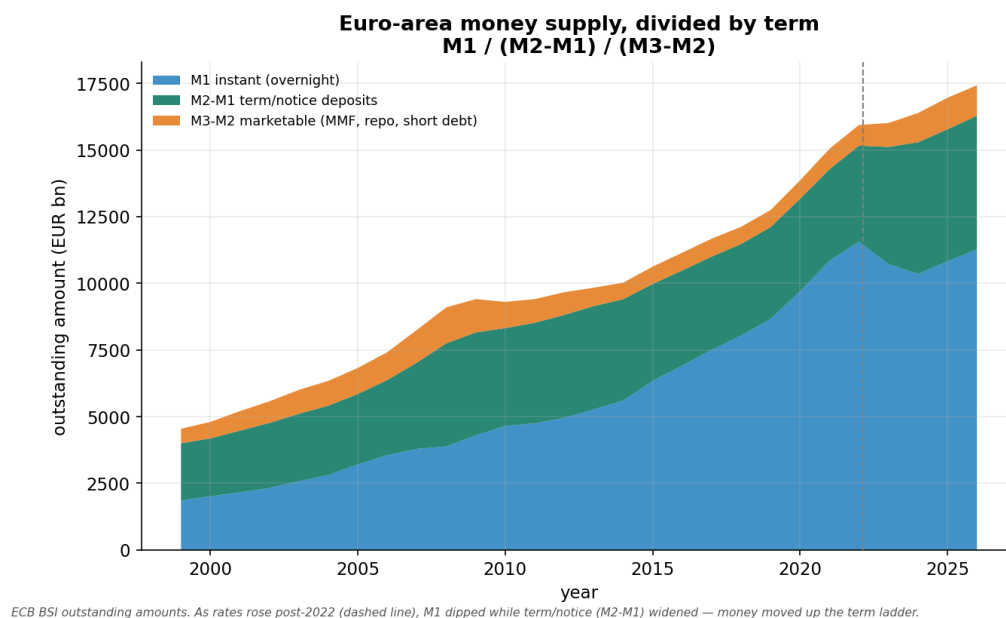


Figure 3: Euro-area money supply decomposed by term: $M1$ (instant), $M2 - M1$ (term/notice deposits), $M3 - M2$ (marketable). The post-2022 widening of the $M2 - M1$ band as $M1$ dips is the term shift.

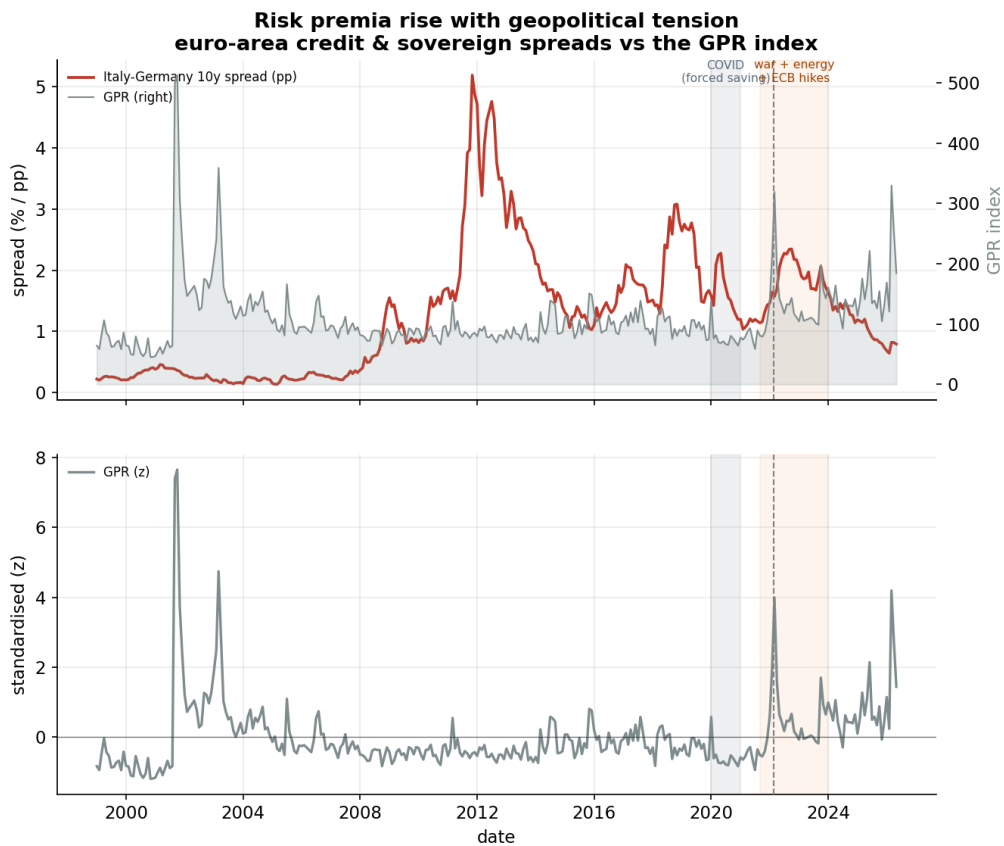
Honest reading. The $M1 \rightarrow$ term migration is exactly what rising rates predict on their own, so this corroborates “the buffer relocated, it did not disappear” but cannot by itself separate precaution from intertemporal substitution.

Table 5: Euro-area money supply by term (EUR bn, annual mean; ECB BSI).

year	M1 instant	M2 – M1 term/notice	M3 – M2 marketable	M3
2019	8,659	3,447	640	12,745
2021	10,846	3,434	766	15,045
2022	11,556	3,604	776	15,936
2023	10,720	4,391	900	16,011
2024	10,351	4,938	1,096	16,384

5 Risk premia and geopolitical tension

The feedback asked whether risk premia rise with geopolitical tension, and whether we can observe it. Three euro-area premia are free, observable, and can be set against the Geopolitical Risk (GPR) index the project already pulls: the *credit* premium (euro high-yield option-adjusted spread); the *sovereign / fragmentation* premium (the Italy–Germany 10-year spread, the cleanest euro-area gauge); and *risk aversion* (the VIX, with VSTOXX as the euro analogue). The script overlays each on GPR, reports the co-movement in levels and monthly changes, and event-windows the February 2022 invasion.



Euro-area risk premia vs the GPR index. Spreads/implied vol jump with geopolitical tension (Feb-2022 line); shaded = COVID and the energy/hiking window.

Figure 4: Euro-area credit and sovereign spreads (top) and VIX (bottom) against the GPR index. Premia rise and jump with geopolitical tension.

Status. The figure shows the *sovereign* (Italy–Germany) spread, pulled from the ECB Data Portal — a fallback that renders this gauge even when FRED is unreachable. The BTP–Bund

spread jumped about +0.6 pp at the February 2022 invasion; over the full sample its *level* correlation with GPR is muted because the 2011–12 euro debt crisis dominates the series, so the event-window jump is the cleaner read. Running `python3 risk_premia.py` with FRED access adds the credit spread (euro high-yield OAS) and the VIX, which show clear positive GPR co-movement and Feb-2022 jumps. Either way the point holds: geopolitical tension is observably priced — an *enabling* fact (the uncertainty is real and measurable in markets) rather than direct evidence on the household saving motive.

6 A forward-looking lens

The core analysis relates the *realised* saving rate to *contemporaneous* uncertainty. A forward-looking version asks whether what households and markets *expect* carries information about saving. We build a caution composite from variables that are, by construction, about the future — expected unemployment and intended saving (Eurostat EI_BSCO_M), and implied volatility (VIX) when FRED is reachable — as the mean of their z-scores, and test it against the realised saving rate (Figures 5–6).

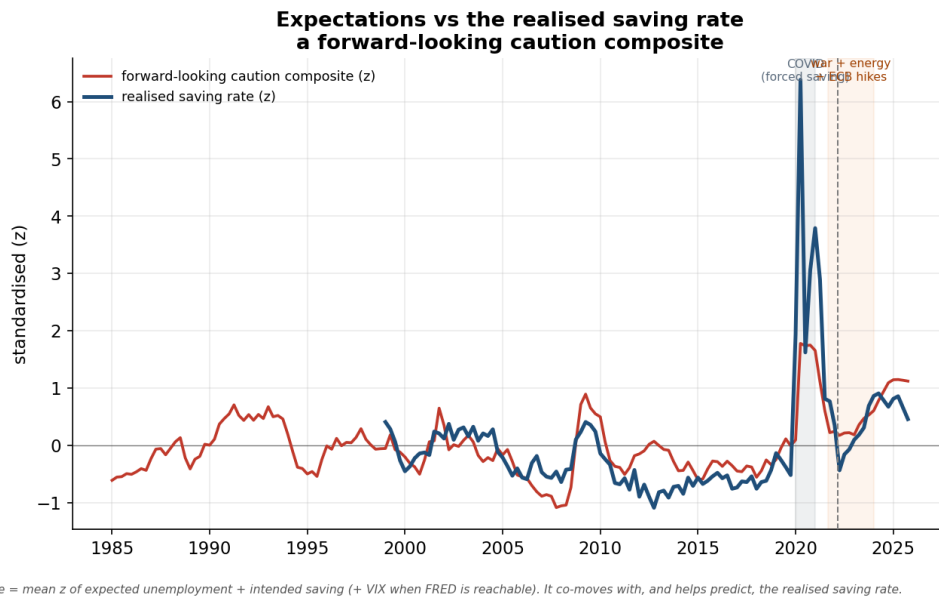


Figure 5: The forward-looking caution composite (z) against the realised saving rate (z).

Result. A one-quarter-ahead regression, $\text{saving}_t = \alpha + \beta \text{composite}_{t-1}$, gives $\beta = +1.88$ ($p < 0.001$, $R^2 = 0.38$, $n = 107$), whereas the *contemporaneous* GPR benchmark explains essentially nothing ($R^2 \approx 0.00$). Two honest caveats. First, the lead-lag correlation peaks *contemporaneously* (+0.76 at $k = 0$), so the right claim is “an already-observable composite predicts saving far better than GPR,” not “expectations strongly lead saving.” Second, the composite includes *intended* saving, which is close to the target by construction; the cleaner forward signal is *expected unemployment* (and, with FRED, the VIX). The substantive implication for the question is a refinement: the operative uncertainty looks like households’ own perceived *income / labour-market risk*, not the abstract geopolitical index — GPR barely moves household saving.

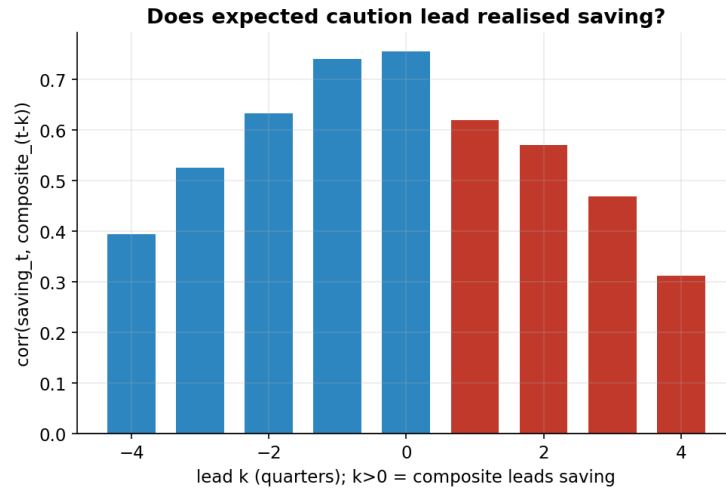


Figure 6: Lead–lag correlation $\text{corr}(\text{saving}_t, \text{composite}_{t-k})$; $k > 0$ means the composite leads.

7 Energy prices and the adequacy of cash

A precautionary buffer is only useful if it can cover a *non-deferrable* shock; the energy bill is the cleanest example. Two findings (Figures 7–8). First, the euro-area energy price level rose about 60% above its 2019 level and stayed elevated, while the instant-cash share of household wealth did *not* rise to meet it (it touched 21.5% in 2022, then fell back to $\sim 18\%$ by 2024) — the non-deferrable bill grew as the cash cushion thinned, and households were moving *out* of cash (§3).

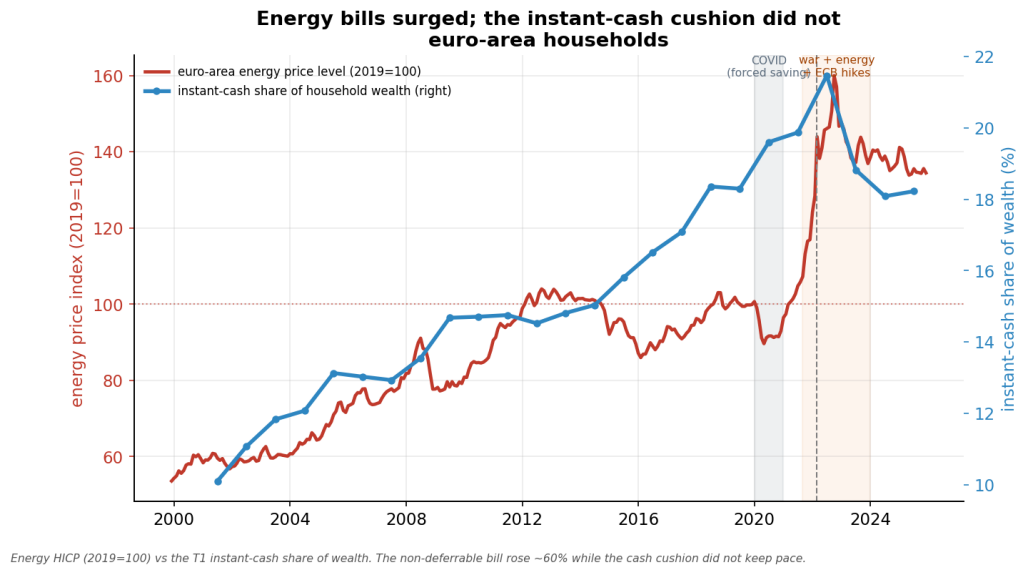


Figure 7: The euro-area energy price level (left, 2019=100) against the instant-cash share of household wealth (right).

Second, the squeeze is *regressive* (Table 6): the lowest-income quintile spends the largest budget share on energy yet *dissaves* (a negative saving rate), while the top quintile saves heavily on the smallest energy share. The aggregate “precautionary buffer” is therefore a top-of-distribution phenomenon, and the households most exposed to an energy shock are the least able to self-insure — they have no buffer to draw on, cash or otherwise.

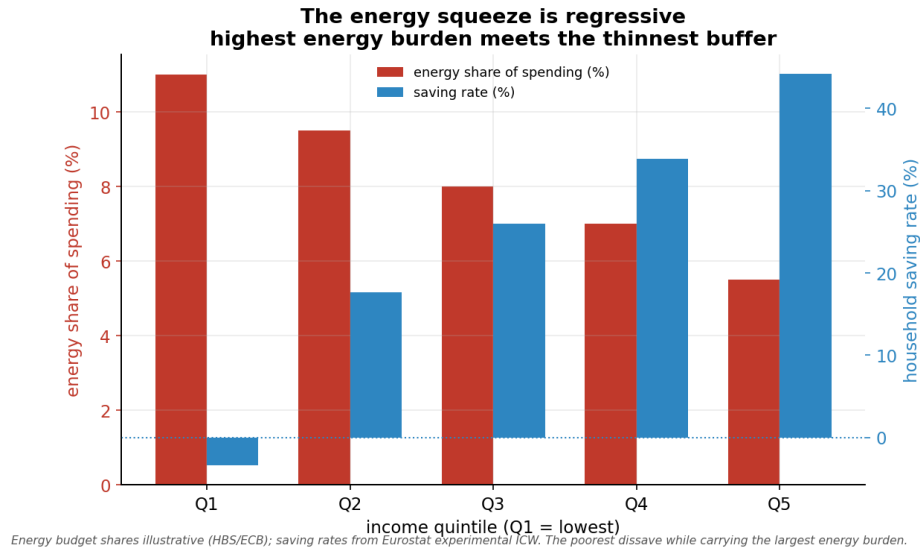


Figure 8: Energy budget share (illustrative) versus the saving rate, by income quintile. Highest burden meets the thinnest buffer.

Table 6: Energy burden and saving capacity by income quintile (Q1 = lowest). Energy shares are illustrative/stylised — verify against the Household Budget Survey; saving rates are from Eurostat experimental ICW.

	Q1	Q2	Q3	Q4	Q5
energy share of spending (%)	11.0	9.5	8.0	7.0	5.5
saving rate (%)	-3.4	+17.6	+26.0	+33.9	+44.2

A caveat to the reframe. “Sellable fast” is not “safe to spend now.” Liquidating marketable (T3) assets to pay an energy bill in a stressed, high-tension state crystallises losses precisely when those asset prices are depressed and risk premia (§5) are elevated. Liquidity in normal times is not the same as adequacy in a crisis — and the most vulnerable households cannot do even this.

8 United States comparison

Does the same picture hold across the Atlantic? We rebuild the liquidity ladder for US households from the Federal Reserve Financial Accounts (Z.1) and set it against the euro area (Figure 9), and contrast the saving rates (Figure 10). The expected contrast: US households hold far *less* in cash and deposits (T1/T2) and far *more* in marketable equities and investment-fund shares (T3) than euro-area households. Their precautionary buffer is therefore even more “sellable-fast” — but also more exposed to *market price risk*, which sharpens the Section 7 caveat: a buffer held in equities can fall in value exactly when it is needed. The US personal saving rate is also structurally lower and more volatile than the euro-area household rate.

figure not found — run `bash run.sh` (this one needs FRED) to generate it.

Figure 9: Household financial wealth by liquidity tier, US vs euro area (stocks, latest). US: Fed Z.1 (households & nonprofits); EA: Eurostat.

figure not found — run `bash run.sh` (this one needs FRED) to generate it.

Figure 10: Saving rates: US (personal, *net*) vs euro area (household, *gross*) — compare the dynamics, not the levels.

Status. `us_comparison.py` is FRED-sourced (Z.1 + the US saving rate), so the two figures regenerate when it is run where FRED is reachable. The US Z.1 series ids are best-effort and the script prints which resolved; shares are taken against *total* financial assets, so an unresolved component shows as a residual rather than inflating the liquid share.

9 What this says about the question

1. **The false binary dissolves.** The post-2022 saving is best read as a *liquid precautionary buffer that also earns yield*. The “precaution versus yield-chasing” dichotomy was an artefact of treating only cash as liquid; on the ladder the broad sellable-fast share of saving flows *rose* (54% → 73%), and about half of household wealth is sellable within days (§3, §4).
2. **The operative uncertainty is income fear, not geopolitics.** Households’ forward-looking expectations predict the saving rate ($R^2 \approx 0.38$ one quarter ahead) while the geopolitical index does not ($R^2 \approx 0$) — so the precautionary channel runs through perceived job/income risk (§6). Risk premia confirm that tension is priced (§5), but that is an enabling fact, not the household motive.
3. **The buffer is unevenly held.** The aggregate masks that the most energy-exposed, most income-fearful households have no cushion at all (§7).

The verdict therefore moves from the presented version’s cautious “probably yield-chasing” to a sharper and better-defended statement: *a liquid precautionary buffer, optimised for yield once rates turned positive, driven more by income fear than by geopolitics, and concentrated among the households who can afford it.*

10 Limitations

- **Correlation, not causation.** Rates, inflation, and uncertainty all moved together after 2022; none of these descriptive results isolates the precautionary channel, and the main project’s formal cross-country / IRF tests still return nulls.
- **Rate-consistency.** The money-supply term shift (§4) is equally consistent with a pure interest-rate story.
- **Mechanical component.** Part of the forward-looking R^2 reflects that the composite includes intended saving (§6).
- **Lower-bound liquidity.** Where Eurostat does not split F5, the equity / funds block is treated as illiquid, so the broad-liquid shares are lower bounds (§2).
- **Illustrative energy shares.** The by-income energy budget shares are stylised and should be verified against the Household Budget Survey.
- **Pending data.** The two US-comparison figures (and the credit / VIX series within the risk-premia figure) regenerate on a machine with FRED access; the risk-premia sovereign

spread already renders via the ECB fallback. All series are pulled live and revised — regenerate before citing.